

SolarWinds Supply Chain Attack

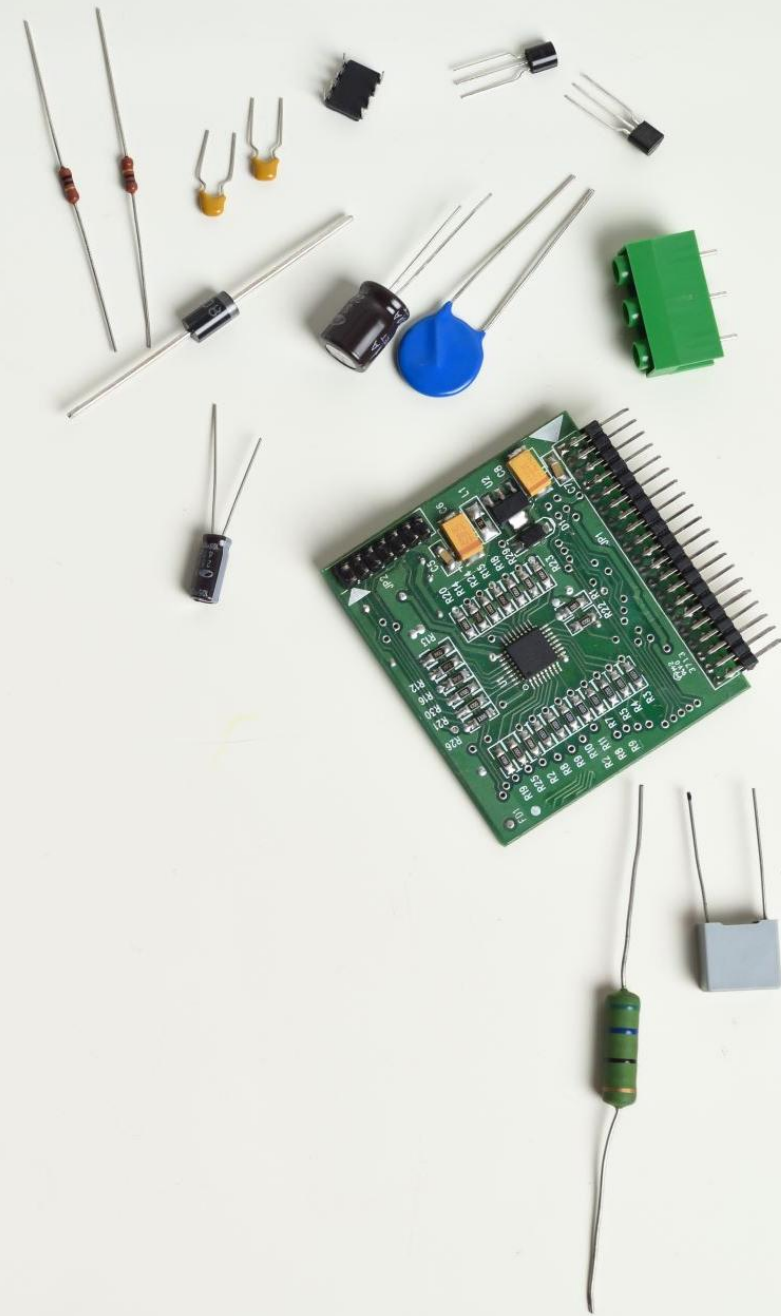
Connie Ha

Computing Security and Technology 2nd year

After watching a movie, I usually want to read the book it was based on to compare them!

Who was hacked?

- SolarWinds is an American IT company that develops software for companies to help them with their network infrastructure.
- They were founded in 1999 and their headquarters is located in Austin, Texas.





What was stolen?

- Sensitive data was exposed that included customer information (credit card info, etc.), government secrets, and intellectual property
- The attackers also gained access to systems that could affect national security, financial stability, and public trust
- Some victims that were affected by this attack included US Commerce and Treasury Departments, the Department of Homeland Security, the National Institutes of Health, State Department, Energy Department and National Nuclear Security Administration

Where did the hack occur?

- The hack occurred within SolarWinds' internal systems that controlled software build and update distribution which was called the SolarWinds Orion software.

```
mirror_mod = modifier_ob.  
# Add mirror object to mirror  
mirror_mod.mirror_object =  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
  
# Selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
  
print("please select exactly  
  
-- OPERATOR CLASSES ----  
  
types.Operator):  
on X mirror to the selected  
object.mirror_mirror_x"  
mirror X"  
  
context):  
context.active_object is not
```

When did the hack occur?

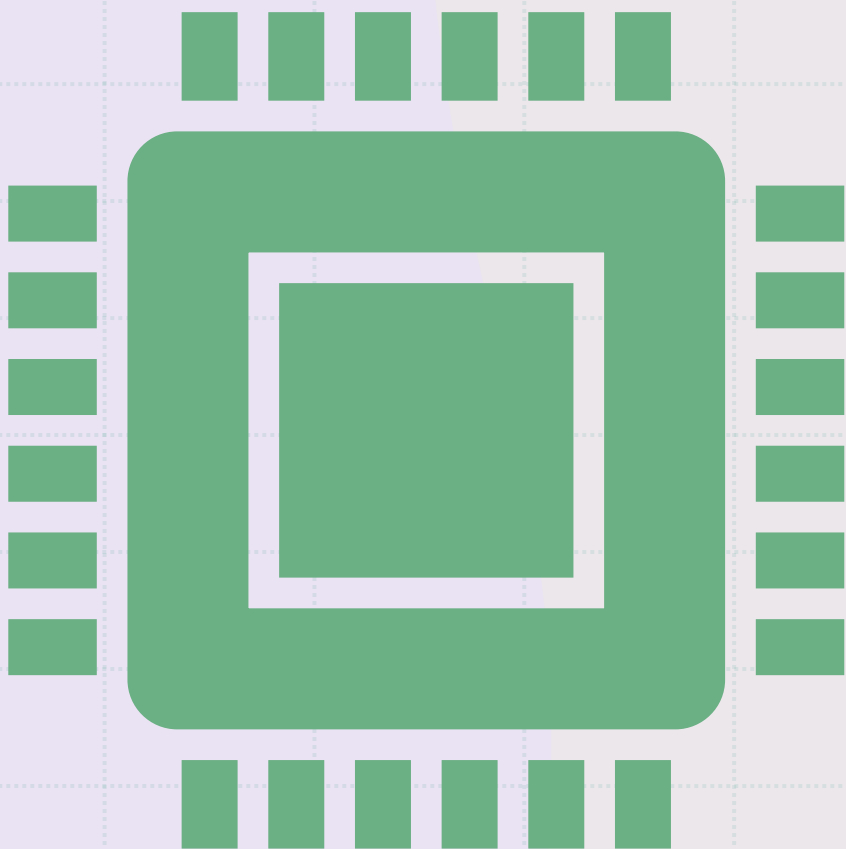
- The attack was not immediately detected, and it was concluded that the attackers had access to the networks for months.
- The exact date it started is not certain, but it is likely to have started in early 2020



Why were they targeted?

- SolarWinds was targeted because they help facilitate network infrastructure for multinational companies and government agencies and therefore the hackers can get into the systems of numerous companies at once
- They are perfect for a supply chain attack





How were they hacked?

- The hackers used a supply chain attack to install malicious software into the Orion Platform
- The Orion Platform allowed for the hackers to create a backdoor to which they were able to access the accounts of users of different organizations
- A Supply chain attack was used because SolarWinds is a third party that has access to a line of organizations' systems, and this was the best way to hack multiple networks indirectly

Cyber tools used

Sealed software code

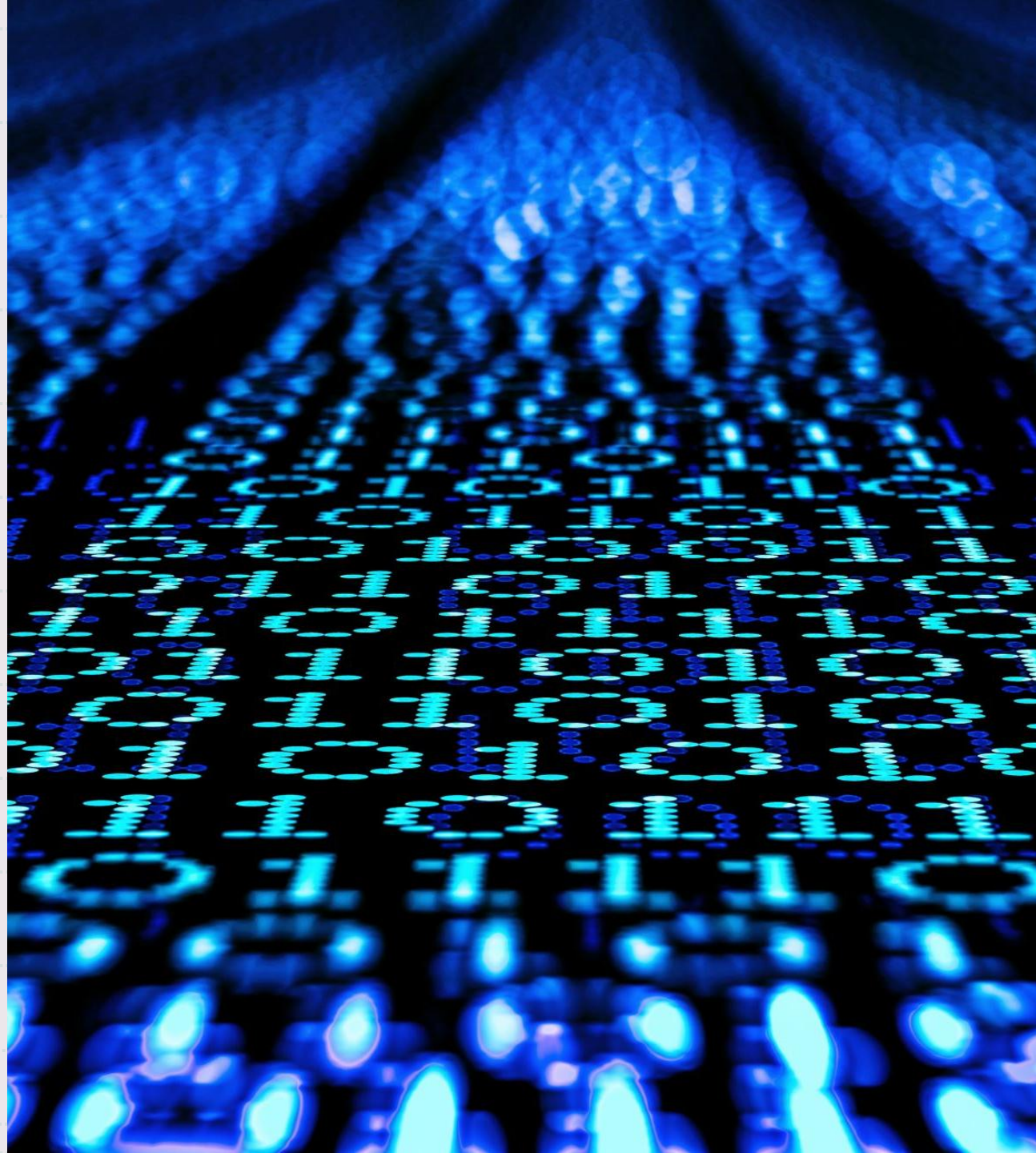
Developed a system that selected targets and mimicked Orion software communication to hide in plain sight using domain names

Code injection

Installed malicious code into the Orion Platform system; the one used was called Sunburst

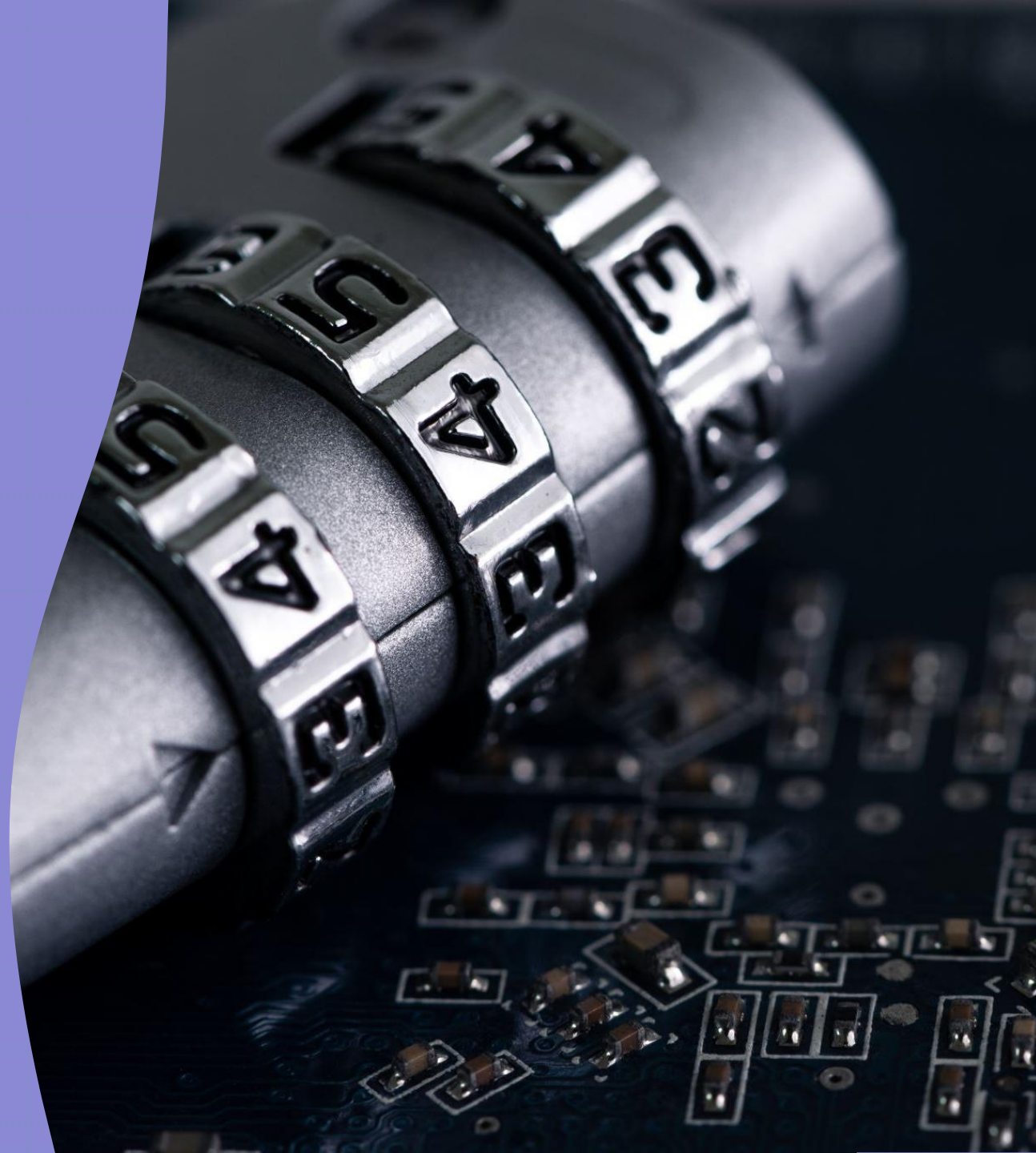
Backdoors

Orion Platform allowed for creation of back door that gave hackers access to numerous organizations' network



Flaws in security exploited

- Their Orion software had insufficient monitoring and unsecure access controls which led to the attack going undetected for months.
- The initial access was likely through a zero-day vulnerability from the third-party software, brute force attack, and a targeted phishing attack



Hacker identification

- It has not been confirmed who was directly involved in this attack, however it is believed by cybersecurity agents that Russia's Foreign Intelligence Service launched a Russian espionage operation that was behind the hack.

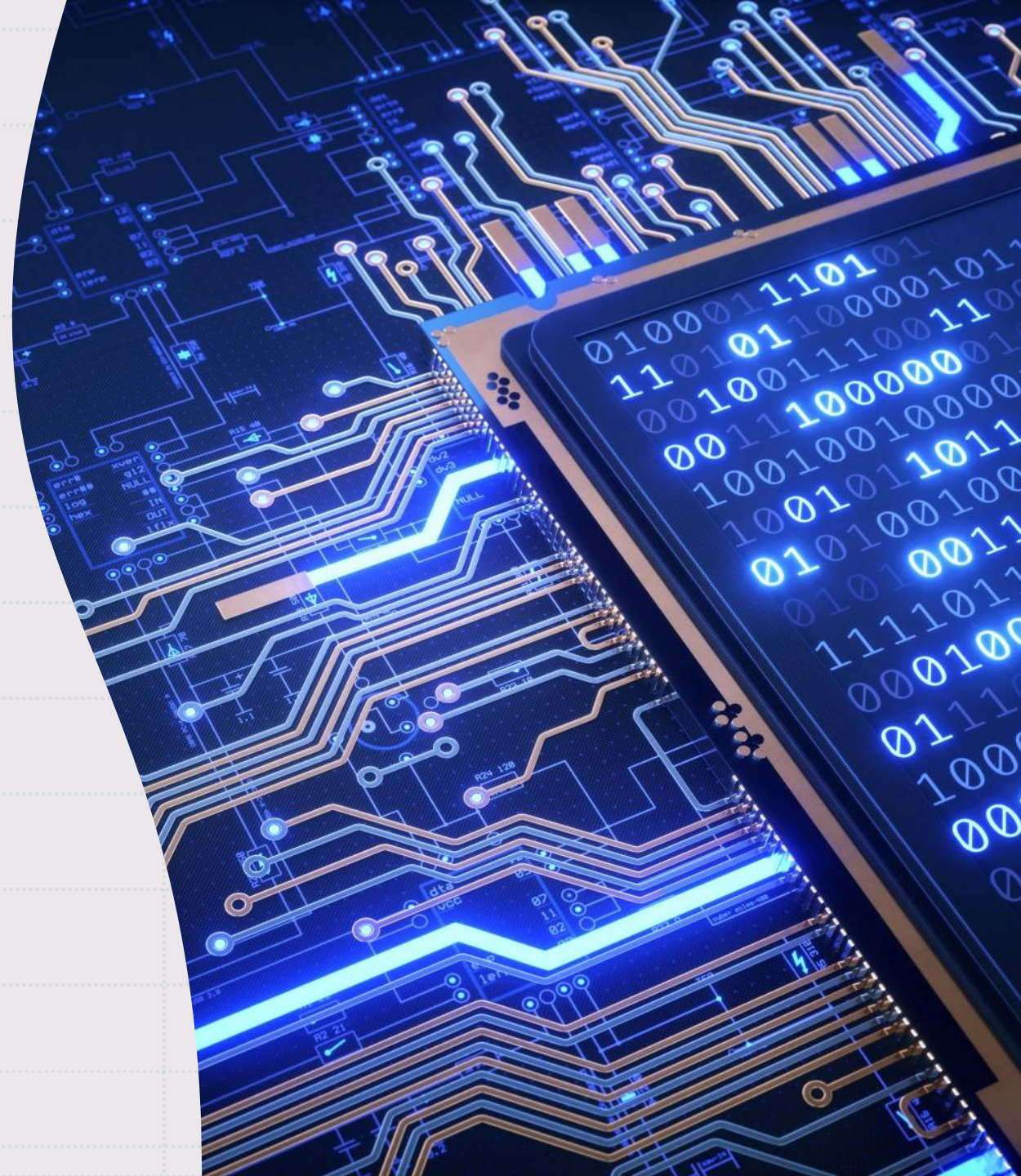


Repercussions of the hack

- The hackers responsible for the SolarWinds attack was able to gain access to sensitive information from companies and government agencies. Highly sensitive information was exposed, which led to a damage in trust in government agencies from the public as well as in SolarWinds as a company.
- SolarWinds underwent many lawsuits and legal liabilities from the organizations they were working with.

Mitigations

- There should be routine security assessments and intrusion detection systems to prevent a future attack from going undetected.
- To ensure a backdoor is not created in the Orion platform system and reduce the chances of future vulnerabilities being exposed, SolarWinds should develop more secure software practice.
- Every organization should have increased cyber security, so they have a fall back in case their third-party software is hacked



Works Cited

- [An Investigative Update of the Cyberattack - Orange Matter \(solarwinds.com\)](#)
- [The SolarWinds hack timeline: Who knew what, and when? | CSO Online](#)
- [SolarWinds Supply Chain Attack Uses SUNBURST Backdoor | Google Cloud Blog](#)
- [SolarWinds hack explained: Everything you need to know \(techtarget.com\)](#)
- [How Russia Used SolarWinds To Hack Microsoft, Intel, Pentagon, Other Networks : NPR](#)
- [SolarWinds tries to rebuild its reputation after hack | Fortune](#)